Rig-less and Blendshape Free Facial Motion Capture via Volume Center Motion Re-targeting

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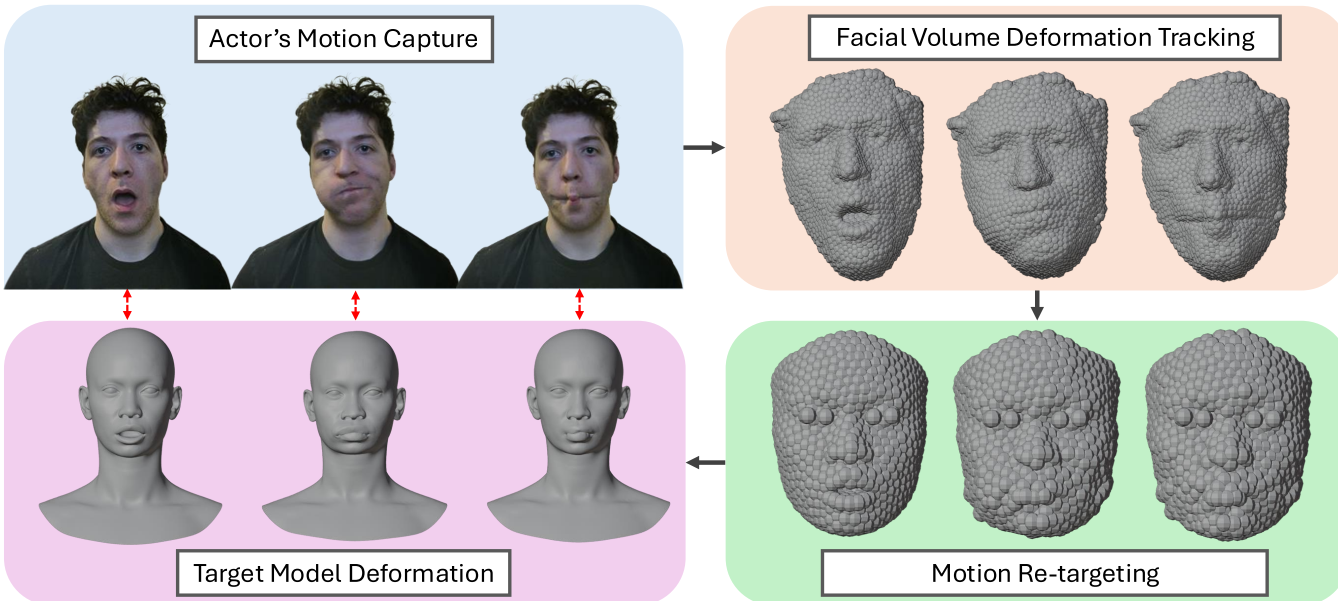

Fig. 1. Overview of our volumetric motion transfer pipeline: 1) Facial performance capture using a single RGB-D device (e.g., iPhone depth sensor or Kinect), followed by depth-to-mesh reconstruction, 2) Dense volume center tracking across the captured sequence, visualized as overlapping spheres that encode local motion and volumetric deformation (facial identity anonymized), and 3) Retargeting of volume center trajectories onto a target head model using anatomical landmark correspondence, followed by volumetric displacement to synthesize animation. Our method transfers facial motion without armatures, blendshapes, or motion markers, while preserving fine-grained deformation and volume consistency.

1 ABSTRACT

Facial motion capture is an essential tool for 3D artists and entertainment studios to create believable animated digital characters. However, existing workflows are expensive and require extensive manual setup; artists must create complex facial rigs using systems of blendshapes and bone-based armatures to deform and manipulate the model. We present Volume Center Motion Re-targeting (VCMR), a novel volumetric approach to facial motion capture and animation. Unlike existing methods that track surface landmarks to solve for and approximate blendshape values, our technique uses volume center tracking to capture an actor's facial deformations and movements. We then establish anatomical correspondences between our actor's captured performance and our target character model and automatically retarget the volume movement onto the volume centers of our target model. Finally, we use the retargetted volume's movement to animate and move the target model's vertices. Our method provides high-density tracking information and eliminates the need for rigging and blendshapes. Experiments are conducted on a variety of target characters, with varying topologies, and multiple motion sequences.

2 INTRODUCTION

Motion capture, the process of transferring motion from one 3D model onto another, remains a foundational yet challenging problem in 3D graphics and animation. While full-body motion capture has matured significantly, high-fidelity facial motion transfer continues to be one of the most labor-intensive and artist-driven components of character production [Zhu and Joslin 2024]. Faces exhibit subtle, highly non-rigid deformations that are difficult to parameterize, yet crucial for conveying emotion, intent, and realism.

Existing facial capture pipelines typically rely on carefully constructed deformation models, such as skeletal rigs [Gall et al. 2009], blendshape bases [Deng et al. 2006], or marker-based tracking systems [Reverdy et al. 2015]. These approaches require extensive manual effort where artists must sculpt expression sets, design control rigs, retopologize geometry, and perform significant cleanup before animation can begin. Even recent AI-driven [Siarohin et al. 2019] or RGB-D–based capture systems [Casas et al. 2016] often assume a predefined deformation space, limiting adaptability across characters and making re-targeting brittle when geometry or topology differs. As a result, high-quality facial animation remains expensive, slow to iterate, and tightly coupled to expert labor.

In this paper, we present VCMR, a fundamentally different approach to facial mocap and re-targeting that eliminates the need for rigs, blendshapes, armatures, or motion markers. Instead of modeling facial motion as transformations of sparse control points or predefined bases, we represent facial dynamics using a dense set of volume centers, tracked over time from raw RGB-D input. These volume centers, conceptually modeled as overlapping spheres, capture local motion, deformation, and volumetric flow across the face.

Our system tracks thousands of volume centers per frame, providing orders of magnitude more motion primitives than traditional rig-based approaches. This redundancy enables robust motion transfer with minimal cleanup while naturally preserving volume and soft-tissue behavior—properties that are often lost in skeletal or blendshape-driven animation. Using anatomical landmark correspondence, we retarget the captured volume center trajectories onto a target mesh and reconstruct facial motion directly through volumetric displacement, without requiring topology-specific preprocessing or optimization.

The entire pipeline operates directly on raw 3D geometry reconstructed from a single consumer-grade smartphone (e.g., iPhone) or an RGB-D camera (e.g., Azure Kinect [Kinect 2015]), enabling fast capture, rapid retargeting, and immediate animation preview. Our experimental evaluations demonstrate that VCMR significantly lowers the barrier to producing high-quality facial animation and enables early-stage performance visualization, rapid iteration, and scalable character creation by removing the dependency on handcrafted deformation models. VCMR, as a volumetric, model-agnostic formulation, opens new directions for facial animation systems that prioritize fidelity, efficiency, and accessibility.

3 BACKGROUND AND RELATED WORK

3.1 3D Face Reconstruction and Performance Capture

Capturing high-quality 3D facial geometry has been a long-standing goal in computer graphics. Early approaches rely on controlled studio setups [Cagniart 2012] using multi-view photogrammetry or structured light [Ha et al. 2015] to obtain highly detailed facial models with accurate geometry, reflectance, and fine-scale surface detail. Multi-camera systems and passive stereo pipelines can achieve sub-millimeter accuracy [Bi et al. 2020], often reconstructing per-frame geometry that is later temporally aligned using optical flow or non-rigid registration techniques. While these methods produce exceptional results, they typically require carefully calibrated camera arrays, controlled lighting, and constrained capture environments, limiting their practicality and scalability.

To reduce capture complexity, recent work has explored the use of low-cost RGB-D sensors [Kinect 2015] for face and body reconstruction [Zollhöfer et al. 2018]. Systems such as KinectFusion [Newcombe et al. 2011] demonstrate real-time reconstruction from commodity depth sensors, but are primarily designed for static or rigid scenes. Subsequent methods extend RGB-D capture to non-rigid subjects by acquiring multiple depth scans from different viewpoints and aligning them using rigid or non-rigid registration [Zollhöfer et al. 2014]. Template-based approaches [Yu et al. 2015] further enable capturing a subject-specific base mesh and continuously deforming it to match the incoming RGB-D scans.

Despite these advances, the primary focus of these systems is on reconstructing geometry over time. The resulting output is typically a temporally aligned mesh sequence that is tightly coupled to a specific subject, topology, or template, making motion reuse and retargeting challenging. In contrast, our work focuses not on reconstruction alone, but on extracting a transferable motion representation that can be applied across different facial geometries.

3.2 Facial Animation and Blendshape-Based Models

Today, one of the most common ways to achieve complex 3D face deformations is through the use of a technique called blendshapes [Cetinaslan and Orvalho 2020; Lewis et al. 2014]. In order to deform a model with blendshapes, an artist begins with a standard neutral position. The artist then creates various deformation positions by moving and manipulating the model's vertices. The 3D software will then interpolate the difference between the vertex movement between the starting base position and the sculpted shape allowing the artist to manipulate the vertices of the model back and forth between these positions. 3D models can have any number of blendshapes and, when they become layered together, artists can achieve very dynamic animation. While they are effective, blendshapes require significant work from an artist to sculpt and manually create expressions. Current solutions like Metahumans from Unreal Engine [Epic Games 2021], Rokoko [Rokoko Electronics 2020], Apple(AR kit) [Apple 2019], and Google MediaPipe [Lugaresi et al. 2019] are compatible with the use of a standardized set of 52 unique facial blendshapes. Higher fidelity ultra realistic characters for film visual effects can require hundreds of blendshapes to produce convincing motion. Existing solutions to auto-generate blendshapes are not very reliable and typically require significant artist cleanup.

Several methods aim to automate blendshape generation by fitting template models to multi-view scans, monocular video, or RGB-D data [Blanz and Vetter 1999; Cao et al. 2015; Li et al. 2010]. While these approaches reduce manual effort, they still rely on predefined deformation spaces, which inherently approximate the original facial structure. Template fitting can introduce geometric smoothing, loss of subject-specific detail, and texture misalignment across expressions [Ichim et al. 2017; Thies et al. 2018].

Marker-based facial capture systems improve tracking accuracy by observing sparse physical markers placed on the face [Beeler et al. 2011], but require invasive setups and still operate on a limited number of control points. Recent facial tracking systems focus on estimating blendshape coefficients or sparse facial landmarks from RGB or RGB-D input in real time [Cao et al. 2014; Kotsia and Pitas 2006]. These systems are designed primarily for efficient tracking rather than high-fidelity motion transfer, and the reconstructed geometry often lacks the spatial detail and volumetric consistency present in the original scans. Moreover, because these methods operate within a fixed blendshape space, they are fundamentally limited in their ability to capture complex soft-tissue motion and volume-preserving deformations [Lewis et al. 2014].

4 VCMR: Technical Design

VCMR transfers complex non-rigid motion between geometrically and anatomically distinct 3D characters **without** requiring vertex-level correspondence, skeletons, or predefined deformation rigs.

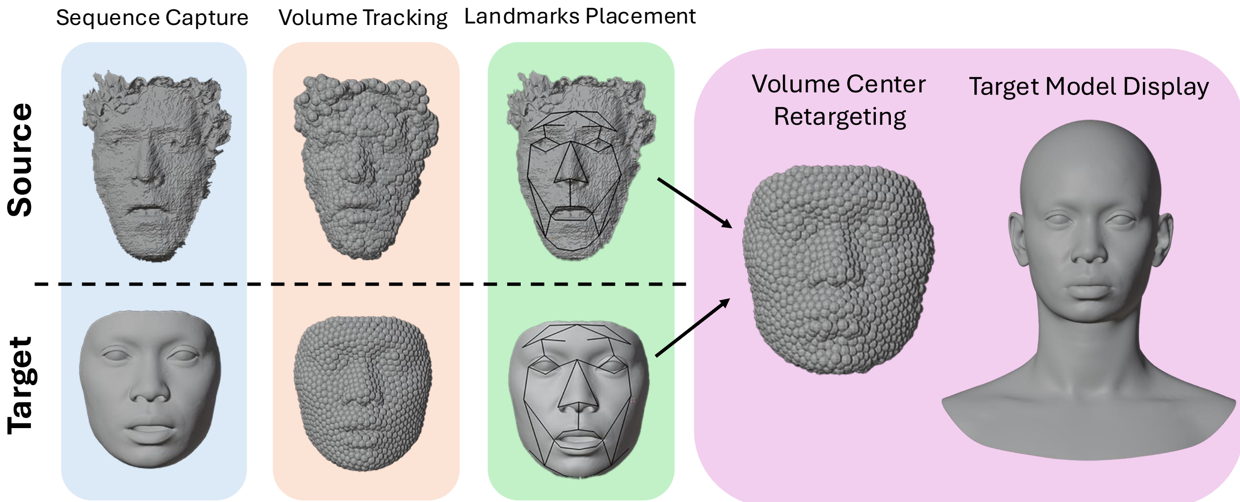

Fig. 2. End-to-end VCMR pipeline. A source performance is captured as a 3D mesh sequence and converted into tracked internal volume centers. Sparse anatomical landmarks defined on a calibration pose provide semantic alignment between source and target. Volumetric motion is then matched and retargeted to the target's volume centers, and propagated to the target mesh to synthesize the animation, without requiring skeletons, or deformation rigs.

Rather than operating on surface vertices, blendshapes, or kinematic chains, VCMR represents motion as the evolution of internal volumetric structure, enabling robust re-targeting across differences in scale, topology, and morphology.

Figure 2 illustrates the end-to-end VCMR pipeline. Given a time-varying 3D scan sequence of a source performance and a static target mesh, VCMR proceeds in five stages: (1) capturing a reference performance as a sequence of meshes, (2) generating and tracking internal volume centers, (3) establishing anatomical correspondence using sparse landmarks, (4) matching and transferring volumetric motion between source and target, and (5) deforming the target mesh to produce the final animation. Each stage is designed to progressively decouple motion from surface geometry while preserving subject-specific deformation characteristics. In the following, we describe each step in detail.

4.1 Capturing the Reference Sequence

Our volume center re-targeting-based motion capture differs heavily from more traditional forms of motion capture. Unlike IMU or image based techniques, VCMR captures actor's movement through a captured sequence of consecutive 3D scanned meshes representing the movement sequence. We then track the actor's volume deformations across the sequence and project this onto our target character, allowing it to capture complex non-rigid deformations such as soft-tissue motion and volume preservation.

In our prototype, the reference sequence is obtained using LiDAR-based 3D scanning at a fixed frame rate, producing a temporally ordered mesh sequence that encodes the full performance. For reliable re-targeting, the sequence includes a calibration frame with minimal self-contact and neutral articulation. For full-body capture, this corresponds to a standard T-pose [Loper et al. 2023]; for facial capture, a neutral expression with an open mouth is used. The target mesh is assumed to be available in a comparable neutral pose, providing a consistent starting configuration for subsequent alignment and correspondence.

4.2 Volume Center Generation and Tracking

Given the reference mesh sequence, we generate a set of internal volume centers that collectively approximate the actor's volumetric structure. These centers act as stable anchors for tracking deformation over time. We compute volume centers using As-Rigid-As-Possible (ARAP) volume tracking [Dvořák et al. 2022], which combines voxelization with fast winding number evaluation to uniformly sample interior regions of the mesh.

ARAP tracking produces a temporally coherent set of volume centers whose trajectories encode both rigid motion and non-rigid deformation. This representation is particularly well-suited for retargeting, as it abstracts motion away from surface tessellation while maintaining spatial coverage of the underlying shape. For visualization and debugging, we represent volume centers as spheres, enabling inspection of tracking quality and detection of drift or instability.

The same volume center generation process is applied to the target mesh, treating it as a single-frame sequence. This ensures consistent volumetric sampling between the source and target, despite differences in geometry or proportions.

4.3 Defining Reference Anatomical Landmarks

Although volume centers provide a stable and geometry-agnostic representation of deformation, they do not encode anatomical semantics. To establish meaningful correspondence between the source and target characters, VCMR augments volumetric structure with a sparse set of anatomical landmarks defined on a calibration pose.

We extract the mesh corresponding to the calibration frame from the reference sequence and use it to define a set of landmark vertices associated with anatomically salient regions. In the current implementation, landmarks are placed manually; however, this step can be automated using learning-based landmark detection and image-based alignment. Effective landmarks capture major anatomical proportions and serve as semantic anchors for normalizing differences in scale, orientation, and morphology between the source and target meshes.

For example, in facial retargeting, landmarks placed at the bridge and tip of the nose allow the system to infer relative nose scale and depth. Similar landmark pairs can be defined around other rigid or semi-rigid structures. When combined with volume center tracking, this sparse semantic information enables motion transfer that preserves subject-specific shape while accurately re-targeting deformation.

Once the landmarks are set, we then extract the position of the volume spheres at the timestamp where the calibration pose occurred and compute an adjacency matrix between the volume center placements and landmark placements. This is accomplished through the following process. We create a 2-dimensional matrix [v * n] where v represents the number of volume centers and n represents the total number of anatomical landmarks. At each value of [v(i), n(j)] we store an integer representing the euclidean distance sum d between volume center n(i) and landmark n(j) derived from the following equation:

$$d = \sqrt{(x_{n(j)} - x_{v(i)})^2 + (y_{n(j)} - y_{v(i)})^2 + (z_{n(j)} - z_{v(i)}))^2} \quad (1)$$

Computing distances relative to aligned anatomical landmarks mitigates differences in scale and orientation between the source and target anatomies. We apply this procedure consistently to both the reference calibration pose and the target mesh, producing comparable volume center–landmark adjacency matrices in each case.

4.4 Matching and Transferring Volume Center Motion

Using the volume center–landmark adjacency matrices constructed in §4.3, we establish correspondence between source and target volume centers by comparing their relative anatomical context. Specifically, we match volume centers based on the similarity of their distance patterns to the shared set of anatomical landmarks.

To enable meaningful comparison across characters with different scales and proportions, we first normalize the landmark–center distance values in each adjacency matrix using z-score normalization. This normalization removes global scale effects and allows correspondence to be determined based on relative spatial relationships rather than absolute distances. We compute the normalized distances as follows:

$$d_{i,k} = \frac{\hat{d}_{i,k} - \mu_i}{\sigma_i + \epsilon} \quad (2)$$

where $\hat{d}_{i,k}$ represents the distance from volume center i to landmark k, and where σ_i and μ_i represent the standard deviation and mean respectively for the data set of values representing the distances between volume center i and each landmark. We use $\epsilon = 10^{-8}$, an arbitrarily small number present to prevent division by zero.

Once we have our values converted to normalized z-scores, we hunt for similar landmark distance matches between the target volume center set, and the source volume center set. We do this by computing a third 2-dimensional matrix of size $[t \times s]$ where t is the number of target volume centers and s is the number of source (reference) volume centers. Then, for each target volume center $t(i)$ and source volume center $s(j)$ pair, we calculate a Pearson correlation coefficient derived from comparing the two volume centers' arrays of normalized distance values. This Pearson correlation coefficient is found using the following equation:

$$r_{i,j} = \frac{\sum_{k=1}^{n}(d_{i,k} - \bar{d}i)(d_{j,k} - \bar{d}j)}{\sqrt{\sum_{k=1}^{n}(d_{i,k} - \bar{d}i)^2 \sum_{k=1}^{n}(d_{j,k} - \bar{d}j)^2}} \quad (3)$$

where $d_{i,k}$ and $d_{j,k}$ represent the normalized distance from the target volume center i and source volume center j corresponding to landmark k, where $\bar{d}i$ and $\bar{d}_j$ represent their respective means, and n represents the total number of utilized landmarks.

The correlation values range from -1 to +1, where +1 indicates perfect positive correlation, 0 indicates no linear relationship, and -1 indicates perfect negative correlation. A correlation threshold is set (in our testing prototype this was set to start at 0.1) that two spheres must exceed in order to be considered a valid match. We than hunt for a fixed value target number (in our testing prototype we looked for 5) of compatible matches that exceed our threshold. If we can't find our target number we lower the threshold and try again until we achieve our goal. If we exceed our target we use the top matches with the highest corelation values.

After we find our matched set of volume spheres, we make sure that they are all well tracked spheres that belong together. In areas like the lips, or other self-contact regions, volume centers can change anatomamical regions and drift between spaces. We must filter these out for our final result. This is done by taking advantage of ARAP volume tracking's affinity weights [Dvořák et al. 2022]. We set an affinity filter to a certain value, (for our test sequences it was at 0.999 but this is dependent on the size of the original tracked meshes) and discarding any spheres that don't meet this requirement. We then assign our selected spheres influence weights for our target sphere, using weight values are found using softmax normalization:

$$w_{i,j} = \frac{\exp(r_{i,j}/\tau)}{\sum_{k=1}^{m}\exp(r_{i,k}/\tau)} \quad (4)$$

where $w_{i,j}$ is the influence weight for source volume center j on target volume center i, $r_{i,j}$ is the correlation between target i and source j, $\tau = 0.2$ is the temperature parameter controlling the sharpness of the weight distribution, and m is the total number of valid source volume centers that are influencing the sphere.

For target volume centers that fail to achieve adequate correlation with any source volume center, we implement a proximity-based fallback system. This system identifies source volume centers within a specified distance threshold (typically 1.0 units) and assigns weights based on inverse distance weighting:

$$w_{i,j} = \frac{1/d_{i,j}}{\sum_{k=1}^{m} 1/d_{i,k}} \quad (5)$$

where $d_{i,j}$ is the normalized distance between target volume center i and source volume center j, and m is the number of nearby source volume centers.

This fallback system enables us to maintain accurate deformations and shapes information across a variety of anatomies and proportions, catching edge cases that our main system may miss.

After matching our target volume centers to sets of source volume centers, we look at the volume center sets positions at frame one and at frame zero. We than combine the delta positions based on the centers weight value we found during the matching process and use this value to offset the target centers to their frame one positions. We than repeat this entire process, recomputing the center's proximity matrix and rematching source and target spheres, for the entire sequence. Rematching volume centers per-frame is important to maintaining accurate motion re-targeting since source volume centers may drift across anatomical regions.

4.5 Deforming the Target Mesh

In the final stage, we synthesize the target animation by propagating the transferred volume center motion to the surface geometry of the target mesh. Each volume center acts as a localized deformation handle whose trajectory encodes both rigid and non-rigid motion derived from the source performance.

Given the target mesh in its calibration pose, we displace its vertices by interpolating the motion of nearby volume centers. For each frame, vertex positions are updated based on a weighted combination of the associated volume center displacements (similar to the displacement field computation from [Chen et al. 2025]), producing smooth, spatially coherent deformation while preserving local shape characteristics. This volumetric formulation avoids the need for explicit skeletons, blendshape rigs, or vertex-level correspondence between the source and target meshes.

Because our method applies deformation over long motion sequences to a single static target model, we incorporate additional temporal smoothing to prevent drift and accumulation of distortion. This smoothing stabilizes vertex trajectories over time while maintaining the fidelity of transferred motion, particularly in regions undergoing repeated or high-frequency deformation.

The resulting sequence of deformed meshes represents the final re-targeted animation and can be exported directly for visualization or downstream use. In practice, we find it effective to represent the generated deformations as blendshapes layered on top of the original target model, enabling integration with standard animation pipelines. In our prototype implementation, we visualize and author the final results using Blender.

5 EVALUATION

5.1 Implementation and Tracked Sequences

All reference performances were captured using an Orbbec Femto Bolt depth camera [Orbbec Inc. 2023] at a fixed frame rate. Processing was performed on a workstation equipped with an NVIDIA RTX 5080 GPU, an AMD Ryzen 9 9900X CPU, and 64 GB of RAM. Motion re-targeting and visualization were implemented as a custom Blender plugin, enabling interactive inspection of volume center trajectories and resulting mesh deformations.

We evaluate VCMR using three facial motion sequences designed to stress different categories of non-rigid deformation: (1) Sequence 1: lip synchronization with speech, (2) Sequence 2: eyebrow motion and eye blinking, and (3) Sequence 3: cheek deformation involving volume compression and expansion. These sequences were selected to capture both localized high-frequency motion and broader volumetric changes. To evaluate robustness across diverse anatomies and geometric styles, we re-target each motion sequence onto three distinct target meshes:

- **MetaHuman (Realistic):** A female MetaHuman model from Unreal Engine, selected to enable direct comparison against MetaHuman Animator [Epic Games 2021], an industry-standard facial capture pipeline.
- **Stylized Elderly Character:** A highly exaggerated elderly male model with non-realistic proportions, used to test robustness to extreme morphology.
- **Stylized Child Character:** A simplified cartoon-style child model, included to evaluate transfer across reduced geometric detail and non-photorealistic anatomy.

Each target mesh is provided in a neutral calibration pose and contains no predefined skeletons, blendshapes, or deformation rigs.

5.2 Results Analysis

Figures 3 compares facial motion transfer results produced by VCMR against Unreal Engine's MetaHuman Animator across multiple target anatomies. We observe a clear difference in the amplitude and character of the resulting motion. Because VCMR represents motion as the aggregated displacement of internal volume centers, the transferred deformation tends to be smoother and less exaggerated than blendshape-driven animation. In contrast, MetaHuman Animator maps observed motion onto a fixed, discrete set of predefined facial expressions, often amplifying motion intensity by snapping to semantically labeled blendshapes rather than reproducing the exact geometric deformation observed in the source performance. While both systems reproduce the overall timing and articulation of speech, MetaHuman exhibits sharper mouth openings and closures corresponding to canonical phoneme shapes. Additionally, our system occasionaly drops some of the faster motions in speech such as quick mouth closures. These are better captured using MetaHuman's pipeline. VCMR preserves more gradual transitions and subtle asymmetries present in the actor's performance, resulting in motion that more closely reflects the continuous deformation observed in the source scans.

The advantage of a volumetric motion representation becomes most apparent in the cheek deformation sequence (Figure 5). Near the end of the sequence, the actor performs a subtle inward lip suction that induces localized volume compression in the cheeks and perioral region. Because VCMR directly transfers internal volume displacement, this compression is faithfully reproduced across all target models, including those with exaggerated or stylized anatomy. Importantly, the ability of VCMR to reproduce such effects generalizes across diverse target geometries. As shown in Figures 3 and 5, volumetric motion transfer remains stable when applied to realistic, stylized, and cartoon characters alike. Despite large differences in facial proportions and surface detail, the transferred motion preserves consistent deformation patterns, demonstrating that VCMR effectively decouples motion representation from surface topology and stylistic detail.

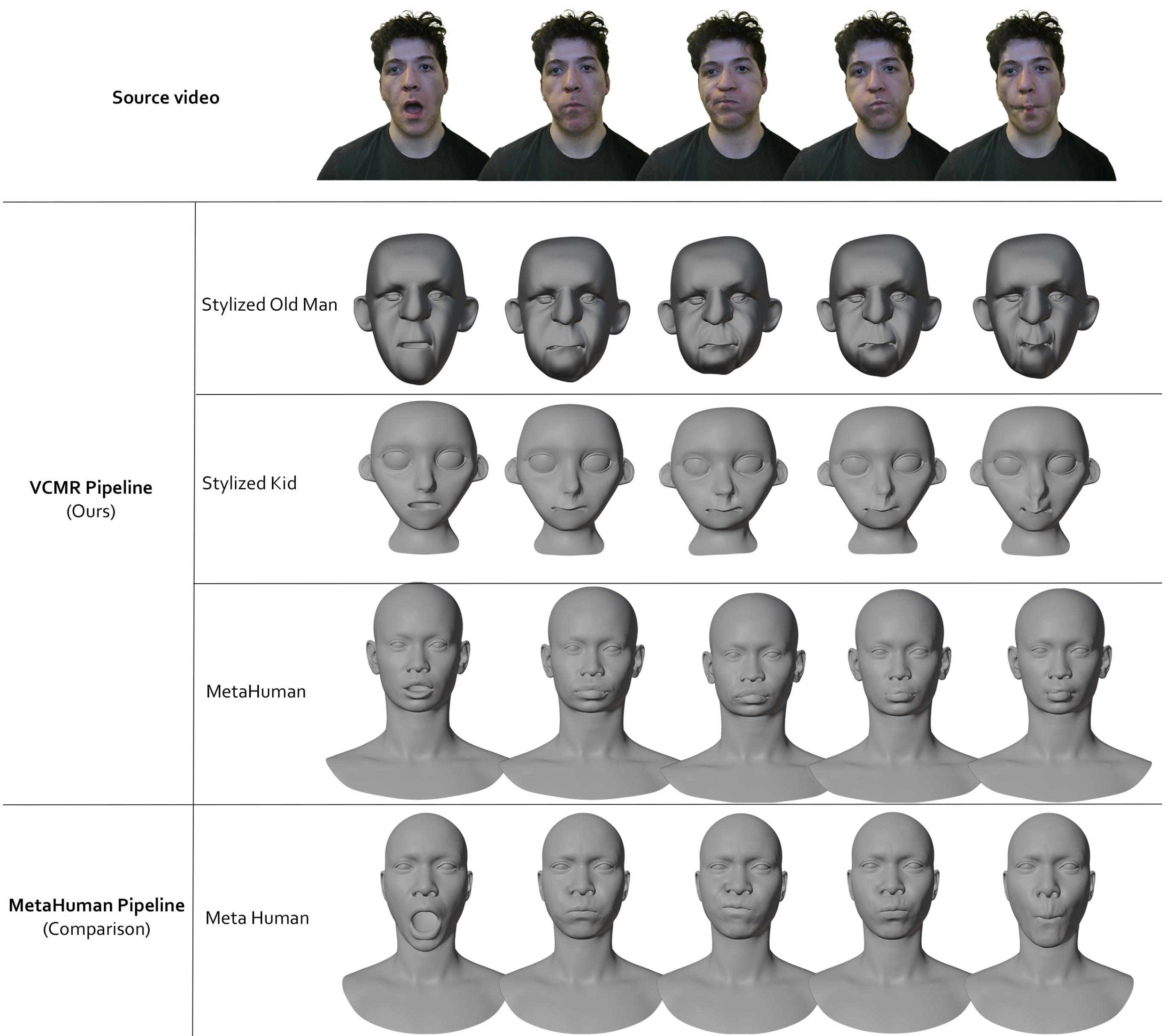

Fig. 3. Lipsync motion transfer comparison across target anatomies on Sequence 1. The top row shows the source video, followed by VCMR on three target meshes with varying anatomy and style. The last row shows retargeted results using MetaHuman Animator as a comparison. MetaHuman Animator relies on discrete, MetaHuman-specific blendshape activations, producing sharper but character-dependent mouth articulations. In contrast, VCMR directly transfers volumetric motion, yielding smoother transitions and subtle asymmetries that naturally generalize to both realistic and stylized target characters.

Eye Blinking and Periorbital Motion. Eye blinking presents a limitation of our current capture setup. Because eyelids are extremely thin structures, they are not reliably reconstructed by the depth sensor used in our experiments. As a result, explicit blinking motion is not faithfully transferred by VCMR.

Despite this limitation, the eyebrow and eye-region sequence (Figure 6) demonstrates that VCMR successfully captures and transfers volumetric motion in the periorbital region. In particular, we observe clear raising and lowering of volume along the supraorbital ridge, as well as coherent deformation within the eye socket region across all target characters. These volumetric changes correspond closely to the actor's brow furrowing and squinting motions, even in frames where eyelid closure itself is not reproduced.

This behavior highlights a key distinction between surface-driven and volume-driven facial animation. While fine-scale surface motions such as eyelid closure are sensor-limited in our current pipeline, broader volumetric deformations surrounding the eyes are robustly

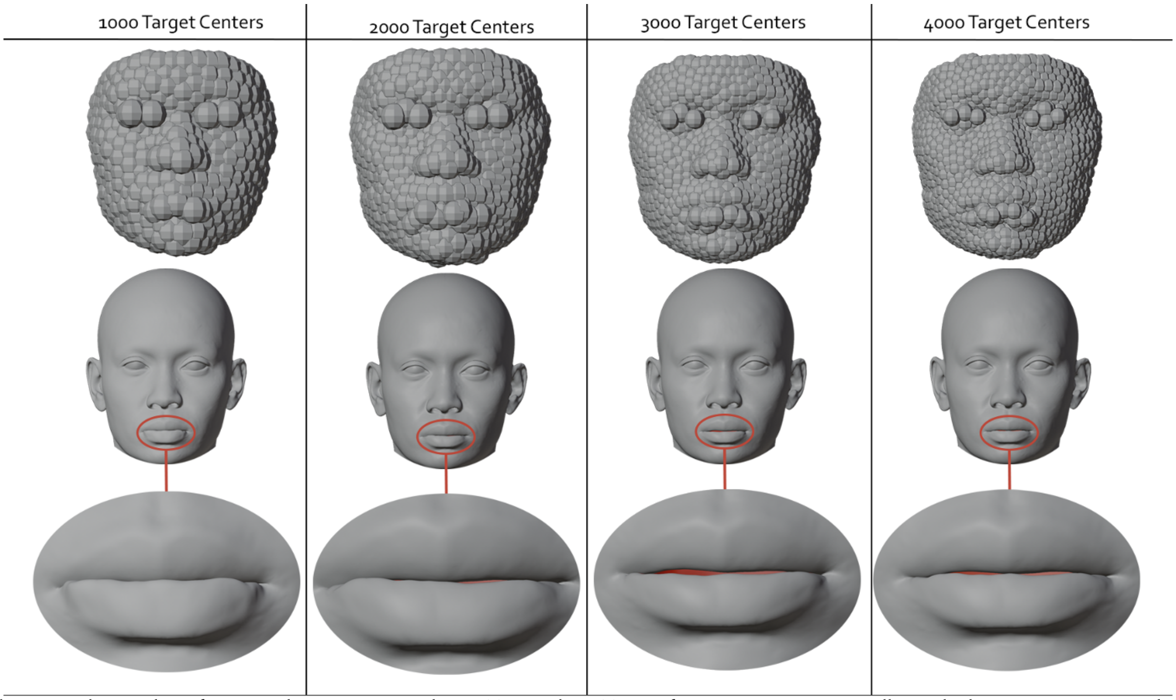

Fig. 4. Ablation on the number of target volume centers in the VCMR pipeline. Motion from Sequence 1, originally tracked using 4000 source volume centers, is re-targeted onto a MetaHuman model using 1000, 2000, 3000, and 4000 target volume centers (left to right). The final frame is shown, where the actor's mouth is intended to be fully closed. The interior of the mouth is highlighted in red to make residual openings and articulation errors clearly visible. Lower target counts produce complete closure but reduced lip detail, while higher target counts preserve local deformation yet exhibit visible red regions, indicating incomplete mouth closure. The 2000-target configuration best suppresses residual interior exposure while maintaining visual fidelity.

captured and transferred. As shown in Figure 6, these effects generalize across realistic, stylized, and cartoon target meshes, indicating that VCMR preserves anatomically meaningful deformation even in the absence of explicit eyelid geometry. We emphasize that this limitation arises from depth sensor resolution rather than from the volumetric re-targeting formulation itself. Incorporating higher-resolution depth capture, multi-view reconstruction, or hybrid RGB-based eyelid tracking would enable explicit modeling of eyelid motion within the same volumetric framework, without requiring changes to the underlying motion transfer algorithm.

5.3 Impact of Target Volume Center Count

We conduct an ablation study to analyze the effect of target volume center count on motion re-targeting quality in VCMR pipeline. The study re-targets Sequence 1, originally tracked using 4000 source volume centers (in Figure 3), onto a MetaHuman model using varying numbers of target volume centers: 1000, 2000, 3000, and 4000. Figure 4 visualizes the final frame of the sequence, where the actor's mouth is intended to be fully closed, the interior of the mouth highlighted for clarity.

The results reveal a clear trade-off between motion fidelity and geometric stability. With lower target counts (1000 and 2000), the mouth closes completely, indicating that a coarser volume representation enforces stronger spatial coupling and suppresses residual gaps. However, at 1000 centers, this comes at the cost of reduced geometric detail, particularly around the lips, where fine-scale deformation is visibly smoothed out. Increasing the target count to 3000 and 4000 improves local deformation fidelity but introduces subtle articulation errors, most notably incomplete mouth closure. This behavior arises from over-parameterization: dense target centers track small, independent volume motions that can accumulate inconsistencies when projected onto a single target mesh, especially in thin and highly articulated regions such as the lips.

6 CONCLUSION

We presented VCMR, a volumetric motion re-targeting framework that transfers complex non-rigid facial motion across geometrically and anatomically diverse 3D characters without requiring vertex correspondence, skeletons, or predefined deformation rigs. VCMR decouples motion capture from mesh topology and enables robust generalization across realistic, stylized, and cartoon characters by representing motion as the evolution of internal volumetric structure rather than surface-level deformations. Our results show that this volumetric formulation preserves subtle performance details, such as localized cheek compression and gradual perioral volume changes, that are difficult to represent using blendshape-based pipelines, while maintaining stable motion transfer across extreme anatomical variation.

References

Apple. 2019. ARKit. https://developer.apple.com/arkit/.

Thabo Beeler, Bernd Bickel, Paul Beardsley, Robert Sumner, and Markus Gross. 2011. High-Quality Single-Shot Capture of Facial Geometry. *ACM Transactions on Graphics* 30, 4 (2011).

Sai Bi, Zexiang Xu, Kalyan Sunkavalli, David Kriegman, and Ravi Ramamoorthi. 2020. Deep 3d capture: Geometry and reflectance from sparse multi-view images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 5960–5969.

Volker Blanz and Thomas Vetter. 1999. A Morphable Model for the Synthesis of 3D Faces. *Proceedings of SIGGRAPH* (1999).

Cedric Cagniart. 2012. *Motion Capture of Deformable Surfaces in Multi-View Studios*. Ph. D. Dissertation. Université de Grenoble.

Chen Cao, Qiming Hou, and Kun Zhou. 2014. FaceWarehouse: A 3D Facial Expression Database for Visual Computing. *IEEE Transactions on Visualization and Computer Graphics* 20, 3 (2014).

Chen Cao, Yanlin Weng, Stephen Lin, and Kun Zhou. 2015. Real-Time High-Fidelity Facial Performance Capture. *ACM Transactions on Graphics* 34, 4 (2015).

Dan Casas, Andrew Feng, Oleg Alexander, Graham Fyffe, Paul Debevec, Ryosuke Ichikari, Hao Li, Kyle Olszewski, Evan Suma, and Ari Shapiro. 2016. Rapid photorealistic blendshape modeling from RGB-D sensors. In *Proceedings of the 29th international conference on computer animation and social agents*. 121–129.

Ozan Cetinaslan and Veronica Orvalho. 2020. Sketching manipulators for localized blendshape editing. *Graphical Models* 108 (2020), 101059.

Guodong Chen, Filip Hácha, Libor Váša, and Mallesham Dasari. 2025. TVMC: Time-Varying Mesh Compression Using Volume-Tracked Reference Meshes. In *Proceedings of the 16th ACM Multimedia Systems Conference*. 79–89.

Zhigang Deng, Pei-Ying Chiang, Pamela Fox, and Ulrich Neumann. 2006. Animating blendshape faces by cross-mapping motion capture data. In *Proceedings of the 2006 symposium on Interactive 3D graphics and games*. 43–48.

Jan Dvořák, Zuzana Káčereková, Petr Vaněček, Lukáš Hruda, and Libor Váša. 2022. As-rigid-as-possible volume tracking for time-varying surfaces. *Computers & Graphics* 102 (2022), 329–338.

Epic Games. 2021. MetaHuman Creator: High-Fidelity Digital Humans in Unreal Engine. https://www.unrealengine.com/metahuman. Online; accessed Jan 2026.

Juergen Gall, Carsten Stoll, Edilson De Aguiar, Christian Theobalt, Bodo Rosenhahn, and Hans-Peter Seidel. 2009. Motion capture using joint skeleton tracking and surface estimation. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*. Ieee, 1746–1753.

Hyowon Ha, Tae-Hyun Oh, and In So Kweon. 2015. A multi-view structured-light system for highly accurate 3D modeling. In *2015 International Conference on 3D Vision*. IEEE, 118–126.

Alexandru Ichim, Sofien Bouaziz, and Mark Pauly. 2017. Dynamic 3D Avatar Creation from Hand-Held Video Input. *ACM Transactions on Graphics* 36, 4 (2017).

Kinect. 2015. Azure Kinect DK. https://azure.microsoft.com/en-us/services/kinect-dk/. Online. Accessed: Dec 2024.

Irene Kotsia and Ioannis Pitas. 2006. Facial expression recognition in image sequences using geometric deformation features and support vector machines. *IEEE transactions on image processing* 16, 1 (2006), 172–187.

John P Lewis, Ken Anjyo, Taehyun Rhee, Mengjie Zhang, Frederic H Pighin, and Zhigang Deng. 2014. Practice and theory of blendshape facial models. *Eurographics (State of the Art Reports)* 1, 8 (2014), 2.

Hao Li, Thibaut Weise, and Mark Pauly. 2010. Example-Based Facial Rigging. *ACM Transactions on Graphics* 29, 4 (2010).

Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J Black. 2023. SMPL: A skinned multi-person linear model. In *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*. 851–866.

Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. *arXiv preprint arXiv:1906.08172* (2019). https://arxiv.org/abs/1906.08172

Richard A Newcombe, Shahram Izadi, Otmar Hilliges, David Molyneaux, David Kim, Andrew J Davison, Pushmeet Kohi, Jamie Shotton, Steve Hodges, and Andrew Fitzgibbon. 2011. Kinectfusion: Real-time dense surface mapping and tracking. In *2011 10th IEEE international symposium on mixed and augmented reality*. Ieee, 127–136.

Orbbec Inc. 2023. Orbbec Femto Bolt Depth Camera. https://www.orbbec.com/products/femto-bolt. Accessed: Jan 2026.

Clément Reverdy, Sylvie Gibet, and Caroline Larboulette. 2015. Optimal marker set for motion capture of dynamical facial expressions. In *Proceedings of the 8th ACM SIGGRAPH Conference on Motion in Games*. 31–36.

Rokoko Electronics. 2020. Rokoko: Real-Time Motion Capture for Digital Humans. https://www.rokoko.com. Online; accessed Jan 2026.

Aliaksandr Siarohin, Stéphane Lathuilière, Sergey Tulyakov, Elisa Ricci, and Nicu Sebe. 2019. Animating arbitrary objects via deep motion transfer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2377–2386.

Justus Thies, Michael Zollhöfer, Marc Stamminger, Christian Theobalt, and Matthias Nießner. 2018. Face2Face: real-time face capture and reenactment of RGB videos. *Commun. ACM* 62, 1 (Dec. 2018), 96–104. https://doi.org/10.1145/3292039

Rui Yu, Chris Russell, Neill DF Campbell, and Lourdes Agapito. 2015. Direct, dense, and deformable: Template-based non-rigid 3d reconstruction from rgb video. In *Proceedings of the IEEE international conference on computer vision*. 918–926.

ChangAn Zhu and Chris Joslin. 2024. A review of motion retargeting techniques for 3D character facial animation. *Computers Graphics* 123 (2024), 104037. https://doi.org/10.1016/j.cag.2024.104037

Michael Zollhöfer, Matthias Nießner, Shahram Izadi, Christoph Rehmann, Christopher Zach, Matthew Fisher, Chenglei Wu, Andrew Fitzgibbon, Charles Loop, Christian Theobalt, et al. 2014. Real-time non-rigid reconstruction using an RGB-D camera. *ACM Transactions on Graphics (ToG)* 33, 4 (2014), 1–12.

Michael Zollhöfer, Patrick Stotko, Andreas Görlitz, Christian Theobalt, Matthias Nießner, Reinhard Klein, and Andreas Kolb. 2018. State of the art on 3D reconstruction with RGB-D cameras. In *Computer graphics forum*, Vol. 37. Wiley Online Library, 625–652.

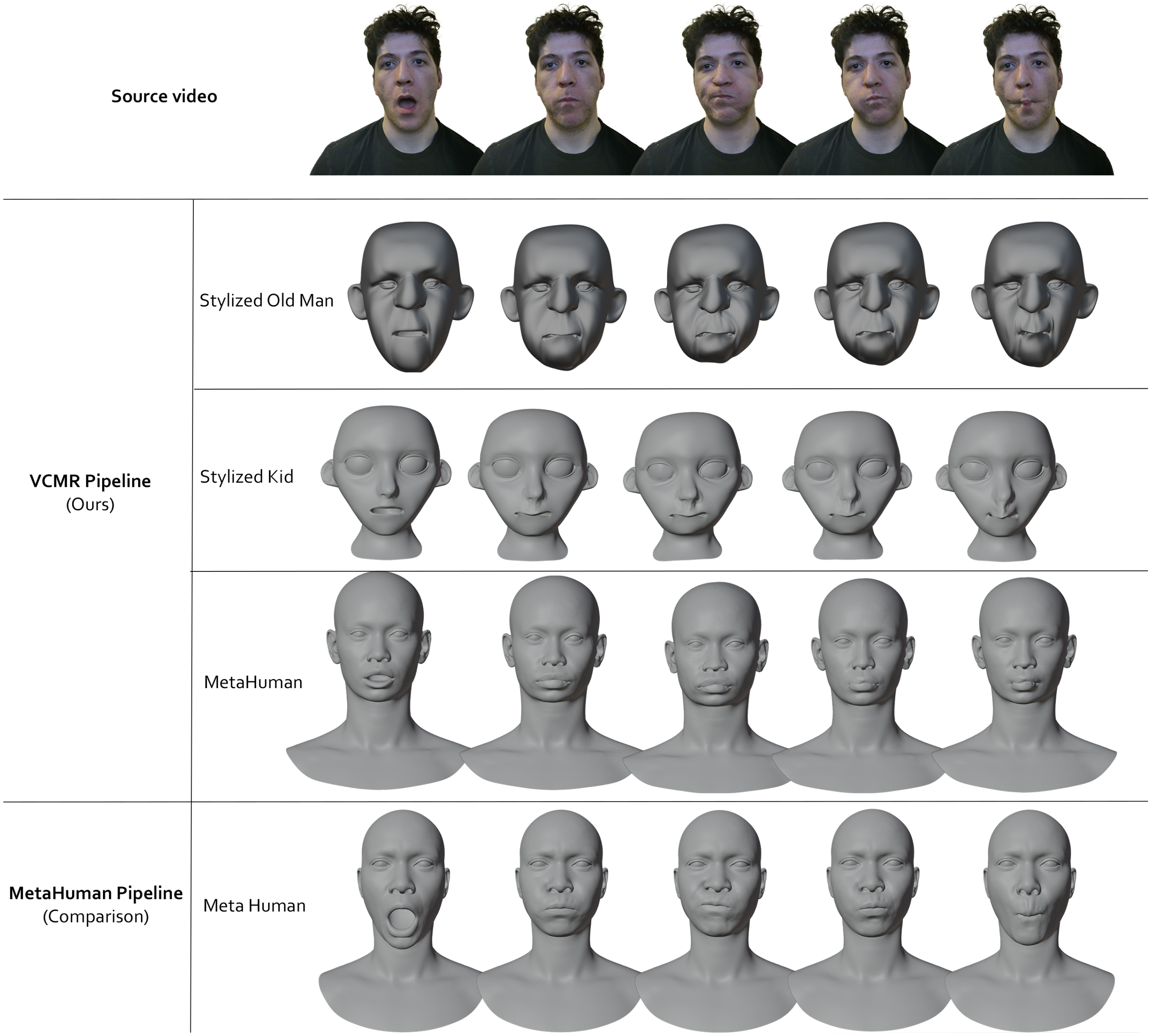

Fig. 5. Cheek deformation transfer across target anatomies on Sequence 2. The top row shows the source video, followed by VCMR on three target meshes, and the retargeted results using MetaHuman Animator for comparison. The sequence emphasizes cheek inflation and compression during speech. VCMR transfers cheek volume changes consistently across diverse facial geometries, preserving smooth, continuous deformation without relying on predefined expression templates.

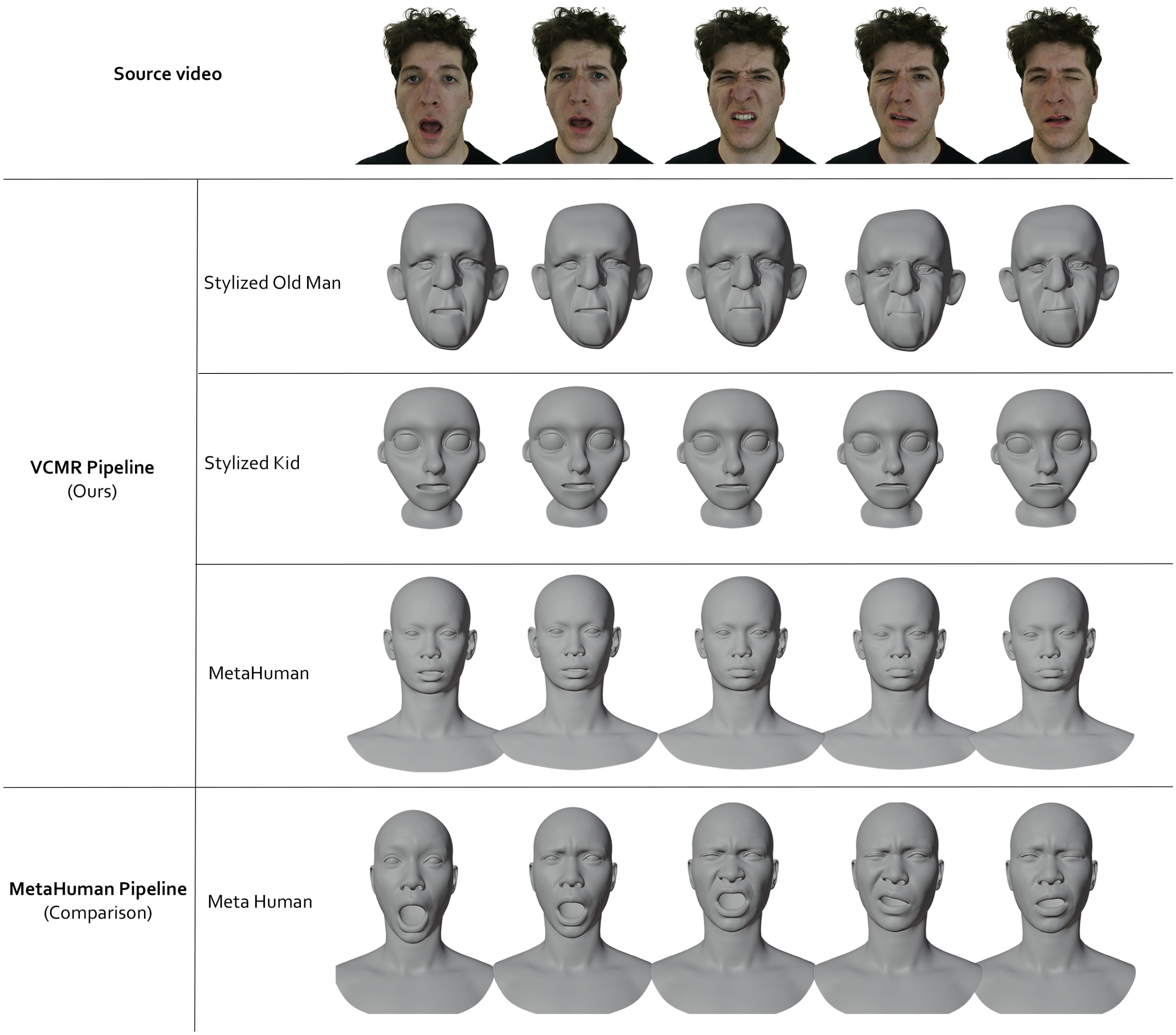

Fig. 6. Eye and eyebrow motion transfer across target anatomies on Sequence 3. The top row shows the source video, followed by VCMR on three target meshes, and the retargeted results using MetaHuman Animator for comparison. While explicit eyelid closure is not captured due to depth sensor limitations, VCMR successfully transfers volumetric motion in the periorbital region, including eyebrow elevation, brow furrowing, and coherent deformation around the eye socket. These effects generalize consistently across realistic, stylized, and cartoon characters.